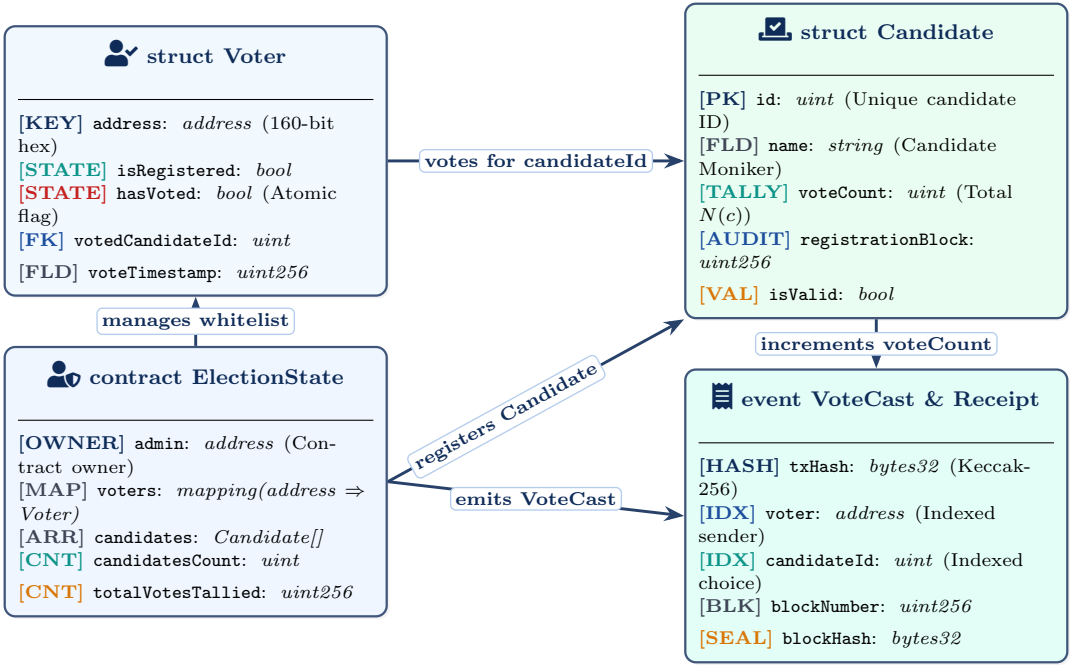

struct Voter
[KEY] address: address (160-bit hex)
[STATE] isRegistered: bool
[STATE] hasVoted: bool (Atomic flag)
[FK] votedCandidateId: uint
[FLD] voteTimestamp: uint256
votes for candidateId
struct Candidate
[PK] id: uint (Unique candidate ID)
[FLD] name: string (Candidate Moniker)
[TALLY] voteCount: uint (Total $N(c)$)
[AUDIT] registrationBlock: uint256
[VAL] isValid: bool
manages whitelist
increments voteCount
contract ElectionState
[OWNER] admin: address (Contract owner)
[MAP] voters: mapping(address ⇒ Voter)
[ARR] candidates: Candidate[]
[CNT] candidatesCount: uint
[CNT] totalVotesTallied: uint256
registers Candidate
emits VoteCast
event VoteCast & Receipt
[HASH] txHash: bytes32 (Keccak-256)
[IDX] voter: address (Indexed sender)
[IDX] candidateId: uint (Indexed choice)
[BLK] blockNumber: uint256
[SEAL] blockHash: bytes32